Ultron

A Fine-Tuned Tool-Calling Agent for Authorized Penetration Testing

The Problem

Pentesters spend significant time translating engagement plans into exact tool invocations. Ultron lets an operator describe a step in plain English and have a model select and correctly format the right tool call — without giving up scope control or human oversight.

Our Approach

We fine-tuned Qwen3.5-2B — chosen over Qwen3-1.7B, Qwen2.5-1.5B, Llama-3.2, Gemma-3-4B, Phi-4-mini, and SmolLM3-3B for its native function calling, best reasoning-per-parameter at the 2B tier, 262K context, and Apache 2.0 license — using QLoRA on a custom tool-call dataset, then exported it as a Q6_K GGUF for fast, offline local inference.

Results So Far

Metric	Value	Metric	Value
Final training loss	1.2061	Steps	30 / 30
Smoothed avg. loss	1.5623	Tokens processed	40,580
Method	QLoRA	Export format	GGUF, Q6_K

Held-out task-accuracy evaluation is planned but not yet run — see Evaluation Report.

Safety by Design

- No tool executes without a registered, authorized engagement scope.
- Medium/high-risk actions require explicit human confirmation before execution.
- Every call, argument, and result is logged for audit.

What's Next

- Run held-out and base-vs-fine-tuned evaluation (Evaluation Report §3).
- Prompt-injection test suite for tool-output handling.
- Expand the custom dataset beyond the initial 40K-token run.

Full documentation: Model Card · Dataset Card · Tool Schema Reference · API Integration Guide · Responsible Use Policy · Evaluation Report · Fine-Tuning Report